

DATABASE MANAGEMENT SYSTEMS

PRACTICAL FILE



ATMA RAM SANATAN DHARMA COLLEGE
UNIVERSITY OF DELHI



SUBMITTED BY
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ROLL NUMBER
24/48013



COURSE
BSC (HONS) COMPUTER SCIENCE



SUBMITTED TO
MR. DHARMENDRA SIR



4TH SEMESTER



Guidelines of B.Sc. (H) Computer Science Semester III

DSC: DATABASE MANAGEMENT SYSTEMS

Unit	Topics	Chapter	Ref.	Suggested Number of Weeks
1	Introduction to Database: Purpose of database system, Characteristics of database approach, data models, database management system, database system architecture, three-schema architecture, components of DBMS, data independence, and file system approach vs database system approach.	1.1 - 1.3, 1.4 - 1.5, 1.6, 1.8, 2.1 - 2.2, 2.3.1, 2.4 -2.4.1, 2.6	1	2
2	Entity Relationship (ER) Modeling: Conceptual data modeling - motivation, entities, entity types, attributes, relationships, relationship types, constraints on relationship, Entity Relationship Diagram notation.	3.1-3.7, 3.9.1	1	2
3	Relational Data Model: Update anomalies, Relational Data Model - Concept of relations, schema-instance distinction, keys, relational integrity constraints, referential integrity and foreign keys, relational algebra operators and queries.	5, 8.1 - 8.3.2, 8.4 (except 8.4.3) 8.5	1	2
4	Structured Query Language (SQL): Querying in SQL, DDL to create database and tables, table constraints, update database-update behaviours, DML, aggregation functions, group by and having clauses, retrieve data from the database, generate and query views. Access and manipulate databases using ODBC. Basic Database administration SQL commands.	6.1-6.4, 7 - 7.1.8 (except 7.1.4), 7.3.1-7.3.2, 7.4	1	4
		2*(Pg. 48), 3*(Listing 3.4)	2	
		11*(Pg. 356-357), 18*(Pg. 532-552)	3	
5	Database Design: Mapping an Entity Relationship (ER) model to relational database, functional dependencies and Normal forms, 1NF, 2NF, 3NF and BCNF decompositions and desirable properties of them.	9.1, 14.1 - 14.5 (up to page 488), Chapter 15 15.1.1 (only Armstrong Axioms without proof and the Closure of X under F to find the primary key)	1	3

6	Data Storage and Indexes: Need of file indexes, file organizations, index structures, single- and multi-level indexing, concurrent execution of transactions, ACID properties	16.5, 17.1.1, 17.2 (up to page 613), 20.1.1- 20.1.3, 20.3	1	2
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References:

1. Elmasri, R., Navathe, B. S. Fundamentals of Database Systems, 7th Edition, Pearson Education, 2015.
2. Krogh, J. W. MySQL Connector/Python Revealed: SQL and NoSQL Data Storage Using MySQL for Python Programmers, Apress, 2018.
3. Murach J. Murach's MySQL, 3rd edition, Pearson, 2019.

PRACTICALS

- I. Create and use the following student-society database schema for a college to answer the given (sample) queries using the standalone SQL editor.

STUDENT	<u>Roll No</u>	StudentName	Course	DOB
	Char(6)	Varchar(20)	Varchar(10)	Date

SOCIETY	<u>SocID</u>	SocName	MentorName	TotalSeats
	Char(6)	Varchar(20)	Varchar(15)	Unsigned int

ENROLLMENT	<u>Roll No</u>	<u>SID</u>	DateOfEnrollment
	Char(6)	Char(6)	Date

CODE:

```
mysql> create table STUDENT(
  -> RollNo CHAR(6) PRIMARY KEY,
  -> StudentName VARCHAR(20),
  -> Course VARCHAR(10),
  -> DOB DATE
  -> );
Query OK, 0 rows affected (0.863 sec)

mysql> create table SOCIETY(
  -> SocID CHAR(6) PRIMARY KEY,
  -> SocName VARCHAR(20),
  -> MentorName VARCHAR(15),
  -> TotalSeats INT UNSIGNED
  -> );
Query OK, 0 rows affected (0.341 sec)

mysql> create table ENROLLMENT(
  -> RollNo CHAR(6),
  -> SID CHAR(6),
  -> DateOfEnrollment DATE,
  -> PRIMARY KEY (RollNo, SID),
  -> FOREIGN KEY (RollNo) REFERENCES STUDENT (RollNo),
  -> FOREIGN KEY (SID) REFERENCES SOCIETY (SocID)
  -> );
Query OK, 0 rows affected (0.888 sec)
```

Table Structures:

```
mysql> DESC STUDENT;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| RollNo | char(6) | NO | PRI | NULL | |
| StudentName | varchar(20) | YES | | NULL | |
| Course | varchar(10) | YES | | NULL | |
| DOB | date | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.289 sec)

mysql> DESC SOCIETY;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| SocID | char(6) | NO | PRI | NULL | |
| SocName | varchar(20) | YES | | NULL | |
| MentorName | varchar(15) | YES | | NULL | |
| TotalSeats | int unsigned | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.106 sec)

mysql> DESC ENROLLMENT;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| RollNo | char(6) | NO | PRI | NULL | |
| SID | char(6) | NO | PRI | NULL | |
| DateOfEnrollment | date | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.041 sec)
```

Inserting Data into Student:

```
mysql> INSERT INTO STUDENT VALUES
-> ('S10001','KRISHNA GUPTA','COMP SCI','2006-08-20'),
-> ('S10002','PRINCE CHOUDHARY','CHEMISTRY','2005-12-24'),
-> ('S10003','KRISH KASHYAP','PHYSICS','2006-11-09'),
-> ('S10004','LOVELEEN KAPIL','COMP SCI','2006-09-11'),
-> ('S10005','MANDEEP CHOUDHARY','CHEMISTRY','2006-02-14'),
-> ('S10006','HARIT SHARMA','MATHS','2006-03-16'),
-> ('S10007','KRISH SALARIA','COMP SCI','2006-08-22'),
-> ('S10008','KRRISH RABUGOTRA','CHEMISTRY','2006-02-17'),
-> ('S10009','HARSHDEEP','PHYSICS','2006-05-04'),
-> ('X10019','KAPIL BHAN','COMP SCI','2005-12-17'),
-> ('Z10029','HARSHVARDHAN MEENA','CHEMISTRY','2004-10-11'),
-> ('S10010','YUVRAJ KAPIL','MATHS','2006-12-24'),
-> ('K10045','RUPESH KUMAR','COMP SCI','2006-03-11'),
-> ('A10089','AMIT DHOUDIYAL','CHEMISTRY','2006-10-10'),
-> ('L50007','SONU YADAV','PHYSICS','2006-04-23');
Query OK, 15 rows affected (0.208 sec)
Records: 15 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM STUDENT;
+-----+-----+-----+-----+
| RollNo | StudentName | Course | DOB |
+-----+-----+-----+-----+
| A10089 | AMIT DHOUDIYAL | CHEMISTRY | 2006-10-10 |
| K10045 | RUPESH KUMAR | COMP SCI | 2006-03-11 |
| L50007 | SONU YADAV | PHYSICS | 2006-04-23 |
| S10001 | KRISHNA GUPTA | COMP SCI | 2006-08-20 |
| S10002 | PRINCE CHOUDHARY | CHEMISTRY | 2005-12-24 |
| S10003 | KRISH KASHYAP | PHYSICS | 2006-11-09 |
| S10004 | LOVELEEN KAPIL | COMP SCI | 2006-09-11 |
| S10005 | MANDEEP CHOUDHARY | CHEMISTRY | 2006-02-14 |
| S10006 | HARIT SHARMA | MATHS | 2006-03-16 |
| S10007 | KRISH SALARIA | COMP SCI | 2006-08-22 |
| S10008 | KRRISH RABUGOTRA | CHEMISTRY | 2006-02-17 |
| S10009 | HARSHDEEP | PHYSICS | 2006-05-04 |
| S10010 | YUVRAJ KAPIL | MATHS | 2006-12-24 |
| X10019 | KAPIL BHAN | COMP SCI | 2005-12-17 |
| Z10029 | HARSHVARDHAN MEENA | CHEMISTRY | 2004-10-11 |
+-----+-----+-----+-----+
```


Inserting Data into Society:

```
mysql> INSERT INTO SOCIETY VALUES
-> ('SOC001','Debating','Pardeep Sharma',10),
-> ('SOC002','Dancing','Ajay Verma',8),
-> ('SOC003','Sashakt','Asha Baloria',12),
-> ('SOC004','NSS','Sudhir Kataria',15),
-> ('SOC005','Coding','Bharat Bhushan',20),
-> ('SOC006','Drama','Aman Gupta',5),
-> ('SOC007','Sports','Yograj Singh',25),
-> ('SOC008','Music','Sanjay Gupta',10);
Query OK, 8 rows affected (0.771 sec)
Records: 8 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM SOCIETY;
+-----+-----+-----+-----+
| SocID | SocName | MentorName | TotalSeats |
+-----+-----+-----+-----+
| SOC001 | Debating | Pardeep Sharma | 10 |
| SOC002 | Dancing | Ajay Verma | 8 |
| SOC003 | Sashakt | Asha Baloria | 12 |
| SOC004 | NSS | Sudhir Kataria | 15 |
| SOC005 | Coding | Bharat Bhushan | 20 |
| SOC006 | Drama | Aman Gupta | 5 |
| SOC007 | Sports | Yograj Singh | 25 |
| SOC008 | Music | Sanjay Gupta | 10 |
+-----+-----+-----+-----+
8 rows in set (0.194 sec)
```

Inserting Data into Enrollment:

```
mysql> INSERT INTO ENROLLMENT VALUES
-> ('S10001','SOC001','2018-01-10'),
-> ('S10001','SOC005','2018-01-15'),
-> ('S10002','SOC002','2018-01-12'),
-> ('S10002','SOC003','2018-01-18'),
-> ('S10003','SOC004','2018-01-20'),
-> ('S10004','SOC001','2018-01-22'),
-> ('S10004','SOC002','2018-01-25'),
-> ('S10005','SOC003','2018-01-28'),
-> ('S10006','SOC005','2018-02-01'),
-> ('S10007','SOC005','2018-02-03'),
-> ('S10007','SOC001','2018-02-05'),
-> ('S10008','SOC002','2018-02-07'),
-> ('S10009','SOC004','2018-02-10'),
-> ('X10019','SOC004','2018-02-12'),
-> ('Z10029','SOC003','2018-02-14'),
-> ('S10010','SOC006','2018-02-16'),
-> ('K10045','SOC005','2018-02-18'),
-> ('A10089','SOC007','2018-02-20'),
-> ('L50007','SOC007','2018-02-22'),
-> ('S10003','SOC007','2018-02-25'),
-> ('S10005','SOC007','2018-02-27');
Query OK, 21 rows affected (0.182 sec)
Records: 21 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM ENROLLMENT;
+-----+-----+-----+
| RollNo | SID | DateOfEnrollment |
+-----+-----+-----+
| A10089 | SOC007 | 2018-02-20 |
| K10045 | SOC005 | 2018-02-18 |
| L50007 | SOC007 | 2018-02-22 |
| S10001 | SOC001 | 2018-01-10 |
| S10001 | SOC005 | 2018-01-15 |
| S10002 | SOC002 | 2018-01-12 |
| S10002 | SOC003 | 2018-01-18 |
| S10003 | SOC004 | 2018-01-20 |
| S10003 | SOC007 | 2018-02-25 |
| S10004 | SOC001 | 2018-01-22 |
| S10004 | SOC002 | 2018-01-25 |
| S10005 | SOC003 | 2018-01-28 |
| S10005 | SOC007 | 2018-02-27 |
| S10006 | SOC005 | 2018-02-01 |
| S10007 | SOC001 | 2018-02-05 |
| S10007 | SOC005 | 2018-02-03 |
| S10008 | SOC002 | 2018-02-07 |
| S10009 | SOC004 | 2018-02-10 |
| S10010 | SOC006 | 2018-02-16 |
| X10019 | SOC004 | 2018-02-12 |
| Z10029 | SOC003 | 2018-02-14 |
+-----+-----+-----+
21 rows in set (0.030 sec)
```

Query 1: Retrieve names of students enrolled in any society.

```
mysql> SELECT DISTINCT s.StudentName from Student s
-> join Enrollment e on s.RollNo = e.RollNo;
+-----+
| StudentName |
+-----+
| AMIT DHOUNDIYAL |
| RUPESH KUMAR |
| SONU YADAV |
| KRISHNA GUPTA |
| PRINCE CHOUDHARY |
| KRISH KASHYAP |
| LOVELEEN KAPIL |
| MANDEEP CHOUDHARY |
| HARIT SHARMA |
| KRISH SALARIA |
| KRRISH RABUGOTRA |
| HARSHDEEP |
| YUVRAJ KAPIL |
| KAPIL BHIAN |
| HARSHVARDHAN MEENA |
+-----+
15 rows in set (0.068 sec)
```

Query 2: Retrieve all society names.

```
mysql> Select SocName from SOCIETY;
+-----+
| SocName |
+-----+
| Debating |
| Dancing |
| Sashakt |
| NSS |
| Coding |
| Drama |
| Sports |
| Music |
+-----+
8 rows in set (0.013 sec)
```

Query 3: Retrieve students' names starting with the letter 'A'.

```
mysql> Select StudentName from STUDENT
-> where StudentName LIKE 'A%';
+-----+
| StudentName |
+-----+
| AMIT DHOUNDIYAL |
+-----+
1 row in set (0.028 sec)
```

Query 4: Retrieve students' details studying in courses 'computer science' or 'chemistry'.

```
mysql> Select * from Student
-> where course in ('COMP SCI','CHEMISTRY');
+-----+-----+-----+-----+
| RollNo | StudentName | Course | DOB |
+-----+-----+-----+-----+
| A10089 | AMIT DHOUDIYAL | CHEMISTRY | 2006-10-10 |
| K10045 | RUPESH KUMAR | COMP SCI | 2006-03-11 |
| S10001 | KRISHNA GUPTA | COMP SCI | 2006-08-20 |
| S10002 | PRINCE CHOUDHARY | CHEMISTRY | 2005-12-24 |
| S10004 | LOVELEEN KAPIL | COMP SCI | 2006-09-11 |
| S10005 | MANDEEP CHOUDHARY | CHEMISTRY | 2006-02-14 |
| S10007 | KRISH SALARIA | COMP SCI | 2006-08-22 |
| S10008 | KRRISH RABUGOTRA | CHEMISTRY | 2006-02-17 |
| X10019 | KAPIL BHAN | COMP SCI | 2005-12-17 |
| Z10029 | HARSHVARDHAN MEENA | CHEMISTRY | 2004-10-11 |
+-----+-----+-----+-----+
10 rows in set (0.019 sec)
```

Query 5: Retrieve students' names whose roll no either starts with 'X' or 'Z' and ends with '9'.

```
mysql> Select StudentName from Student
-> where (RollNo like 'X%' or RollNo like 'Z%')
-> and RollNo Like '%9';
+-----+
| StudentName |
+-----+
| KAPIL BHAN |
| HARSHVARDHAN MEENA |
+-----+
2 rows in set (0.050 sec)
```

Query 6: Find society details with more than N TotalSeats where N is to be input by the user

```
mysql> Select * from society
-> where TotalSeats>5; --Here N=5
+-----+-----+-----+-----+
| SocID | SocName | MentorName | TotalSeats |
+-----+-----+-----+-----+
| SOC001 | Debating | Pardeep Sharma | 10 |
| SOC002 | Dancing | Ajay Verma | 8 |
| SOC003 | Sashakt | Asha Baloria | 12 |
| SOC004 | NSS | Sudhir Kataria | 15 |
| SOC005 | Coding | Bharat Bhushan | 20 |
| SOC007 | Sports | Yograj Singh | 25 |
| SOC008 | Music | Sanjay Gupta | 10 |
+-----+-----+-----+-----+
7 rows in set (0.022 sec)
```


Query 7: Update society table for the mentor name of a specific society

```
mysql> UPDATE SOCIETY
-> SET MentorName = 'Gautam Gambhir'
-> where SocID='SOC007';
Query OK, 1 row affected (0.279 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql> Select * from society;
+-----+-----+-----+-----+
| SocID | SocName | MentorName | TotalSeats |
+-----+-----+-----+-----+
| SOC001 | Debating | Pardeep Sharma | 10 |
| SOC002 | Dancing | Ajay Verma | 8 |
| SOC003 | Sashakt | Asha Baloria | 12 |
| SOC004 | NSS | Sudhir Kataria | 15 |
| SOC005 | Coding | Bharat Bhushan | 20 |
| SOC006 | Drama | Aman Gupta | 5 |
| SOC007 | Sports | Gautam Gambhir | 25 |
| SOC008 | Music | Sanjay Gupta | 10 |
+-----+-----+-----+-----+
8 rows in set (0.012 sec)
```

Query 8: Find society names in which more than five students have enrolled

```
mysql> select SocName from Society s
-> join enrollment e on s.SocID = e.SID
-> group by s.SocID
-> HAVING COUNT(e.RollNo)>5;
Empty set (0.192 sec)
```

Query 9: Find the name of the youngest student enrolled in society 'NSS'

```
mysql> SELECT s.StudentName
-> FROM STUDENT s
-> JOIN ENROLLMENT e ON s.RollNo = e.RollNo
-> JOIN SOCIETY so ON so.SocID = e.SID
-> WHERE so.SocName = 'NSS'
-> AND s.DOB = (
-> SELECT MAX(s2.DOB)
-> FROM STUDENT s2
-> JOIN ENROLLMENT e2 ON s2.RollNo = e2.RollNo
-> JOIN SOCIETY so2 ON so2.SocID = e2.SID
-> WHERE so2.SocName = 'NSS'
-> );
+-----+
| StudentName |
+-----+
| KRISH KASHYAP |
+-----+
1 row in set (0.278 sec)
```

Query 10: Find the name of the most popular society (on the basis of enrolled students)

```
mysql> SELECT SocName
-> FROM SOCIETY s
-> JOIN ENROLLMENT e ON s.SocID = e.SID
-> GROUP BY s.SocID
-> ORDER BY COUNT(*) DESC
-> LIMIT 1;
+-----+
| SocName |
+-----+
| Sports  |
+-----+
1 row in set (0.079 sec)
```

Query 11: Find the name of two least popular societies (on the basis of enrolled students)

```
mysql> SELECT SocName
-> FROM SOCIETY s
-> JOIN ENROLLMENT e ON s.SocID = e.SID
-> GROUP BY s.SocID
-> ORDER BY COUNT(*) ASC
-> LIMIT 2;
+-----+
| SocName |
+-----+
| Drama   |
| Dancing |
+-----+
2 rows in set (0.034 sec)
```

Query 12: Find the students names who are not enrolled in any society.

```
mysql> SELECT StudentName
-> FROM STUDENT
-> WHERE RollNo NOT IN (
->     SELECT RollNo FROM ENROLLMENT
-> );
Empty set (0.032 sec)
```

Query 13: Find the students names enrolled in at least two societies

```
mysql> SELECT s.StudentName
-> FROM STUDENT s
-> JOIN ENROLLMENT e ON s.RollNo = e.RollNo
-> GROUP BY s.RollNo
-> HAVING COUNT(e.SID) >= 2;
+-----+
| StudentName |
+-----+
| KRISHNA GUPTA |
| PRINCE CHOUDHARY |
| KRISH KASHYAP |
| LOVELEEN KAPIL |
| MANDEEP CHOUDHARY |
| KRISH SALARIA |
+-----+
6 rows in set (0.011 sec)
```

Query 14: Find society names in which maximum students are enrolled

```
mysql> SELECT SocName
-> FROM SOCIETY s
-> JOIN ENROLLMENT e ON s.SocID = e.SID
-> GROUP BY s.SocID
-> HAVING COUNT(*) = (
->     SELECT MAX(cnt)
->     FROM (
->         SELECT COUNT(*) AS cnt
->         FROM ENROLLMENT
->         GROUP BY SID
->     ) AS temp
-> );
```

SocName
Coding
Sports

2 rows in set (0.092 sec)

Query 15: Find names of all students who have enrolled in any society and society names in which at least one student has enrolled

```
mysql> SELECT s.StudentName, so.SocName
-> FROM STUDENT s
-> JOIN ENROLLMENT e ON s.RollNo = e.RollNo
-> JOIN SOCIETY so ON so.SocID = e.SID;
```

StudentName	SocName
KRISHNA GUPTA	Debating
LOVELEEN KAPIL	Debating
KRISH SALARIA	Debating
PRINCE CHOUDHARY	Dancing
LOVELEEN KAPIL	Dancing
KRRISH RABUGOTRA	Dancing
PRINCE CHOUDHARY	Sashakt
MANDEEP CHOUDHARY	Sashakt
HARSHVARDHAN MEENA	Sashakt
KRISH KASHYAP	NSS
HARSHDEEP	NSS
KAPIL BHAYAN	NSS
RUPESH KUMAR	Coding
KRISHNA GUPTA	Coding
HARIT SHARMA	Coding
KRISH SALARIA	Coding
YUVRAJ KAPIL	Drama
AMIT DHOUNDIYAL	Sports
SONU YADAV	Sports
KRISH KASHYAP	Sports
MANDEEP CHOUDHARY	Sports

21 rows in set (0.013 sec)

Query 16: Find names of students who are enrolled in any of the three societies 'Debating', 'Dancing' and 'Sashakt'.

```
mysql> SELECT DISTINCT s.StudentName
-> FROM STUDENT s
-> JOIN ENROLLMENT e ON s.RollNo = e.RollNo
-> JOIN SOCIETY so ON so.SocID = e.SID
-> WHERE so.SocName IN ('Debating','Dancing','Sashakt');
```

StudentName
KRISHNA GUPTA
LOVELEEN KAPIL
KRISH SALARIA
PRINCE CHOUDHARY
KRRISH RABUGOTRA
MANDEEP CHOUDHARY
HARSHVARDHAN MEENA

7 rows in set (0.048 sec)

Query 17: Find society names such that its mentor has a name with 'Gupta' in it

```
mysql> SELECT SocName
-> FROM SOCIETY
-> WHERE MentorName LIKE '%Gupta%';
+-----+
| SocName |
+-----+
| Drama   |
| Music   |
+-----+
2 rows in set (0.038 sec)
```

Query 18: Find the society names in which the number of enrolled students is only 10% of its capacity

```
mysql> SELECT s.SocName
-> FROM SOCIETY s
-> JOIN ENROLLMENT e ON s.SocID = e.SID
-> GROUP BY s.SocID
-> HAVING COUNT(e.RollNo) = 0.1 * MAX(s.TotalSeats);
Empty set (0.023 sec)
```

Query 19: Display the vacant seats for each society.

```
mysql> SELECT s.SocName,
-> s.TotalSeats - COUNT(e.RollNo) AS VacantSeats
-> FROM SOCIETY s
-> LEFT JOIN ENROLLMENT e ON s.SocID = e.SID
-> GROUP BY s.SocID;
+-----+-----+
| SocName | VacantSeats |
+-----+-----+
| Debating | 7 |
| Dancing | 5 |
| Sashakt | 9 |
| NSS | 12 |
| Coding | 16 |
| Drama | 4 |
| Sports | 21 |
| Music | 10 |
+-----+-----+
8 rows in set (0.040 sec)
```

Query 20: Increment Total Seats of each society by 10%

```
mysql> UPDATE SOCIETY
-> SET TotalSeats = TotalSeats * 1.10;
Query OK, 8 rows affected (0.094 sec)
Rows matched: 8 Changed: 8 Warnings: 0

mysql> select * from society;
+-----+-----+-----+-----+
| SocID | SocName | MentorName | TotalSeats |
+-----+-----+-----+-----+
| SOC001 | Debating | Pardeep Sharma | 11 |
| SOC002 | Dancing | Ajay Verma | 9 |
| SOC003 | Sashakt | Asha Baloria | 13 |
| SOC004 | NSS | Sudhir Kataria | 17 |
| SOC005 | Coding | Bharat Bhushan | 22 |
| SOC006 | Drama | Aman Gupta | 6 |
| SOC007 | Sports | Gautam Gambhir | 28 |
| SOC008 | Music | Sanjay Gupta | 11 |
+-----+-----+-----+-----+
8 rows in set (0.009 sec)
```

Query 21: Add the enrollment fees paid ('Yes'/'No') field in the enrollment table.

```
mysql> ALTER TABLE ENROLLMENT
-> ADD FeesPaid ENUM('Yes','No');
Query OK, 0 rows affected (0.757 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> select * from enrollment;
```

RollNo	SID	DateOfEnrollment	FeesPaid
A10089	SOC007	2018-02-20	NULL
K10045	SOC005	2018-02-18	NULL
L50007	SOC007	2018-02-22	NULL
S10001	SOC001	2018-01-10	NULL
S10001	SOC005	2018-01-15	NULL
S10002	SOC002	2018-01-12	NULL
S10002	SOC003	2018-01-18	NULL
S10003	SOC004	2018-01-20	NULL
S10003	SOC007	2018-02-25	NULL
S10004	SOC001	2018-01-22	NULL
S10004	SOC002	2018-01-25	NULL
S10005	SOC003	2018-01-28	NULL
S10005	SOC007	2018-02-27	NULL
S10006	SOC005	2018-02-01	NULL
S10007	SOC001	2018-02-05	NULL
S10007	SOC005	2018-02-03	NULL
S10008	SOC002	2018-02-07	NULL
S10009	SOC004	2018-02-10	NULL
S10010	SOC006	2018-02-16	NULL
X10019	SOC004	2018-02-12	NULL
Z10029	SOC003	2018-02-14	NULL

```
21 rows in set (0.024 sec)
```

Query 22: Update date of enrollment of society id 's1' to '2018-01-15', 's2' to the current date and 's3' to '2018-01-02'.

```
mysql> UPDATE ENROLLMENT
-> SET DateOfEnrollment =
-> CASE
->     WHEN SID = 'SOC001' THEN '2018-01-15'
->     WHEN SID = 'SOC002' THEN CURDATE()
->     WHEN SID = 'SOC003' THEN '2018-01-02'
-> END;
Query OK, 21 rows affected (0.077 sec)
Rows matched: 21 Changed: 21 Warnings: 0

mysql> select * from enrollment;
```

RollNo	SID	DateOfEnrollment	FeesPaid
A10089	SOC007	NULL	NULL
K10045	SOC005	NULL	NULL
L50007	SOC007	NULL	NULL
S10001	SOC001	2018-01-15	NULL
S10001	SOC005	NULL	NULL
S10002	SOC002	2026-04-15	NULL
S10002	SOC003	2018-01-02	NULL
S10003	SOC004	NULL	NULL
S10003	SOC007	NULL	NULL
S10004	SOC001	2018-01-15	NULL
S10004	SOC002	2026-04-15	NULL
S10005	SOC003	2018-01-02	NULL
S10005	SOC007	NULL	NULL
S10006	SOC005	NULL	NULL
S10007	SOC001	2018-01-15	NULL
S10007	SOC005	NULL	NULL
S10008	SOC002	2026-04-15	NULL
S10009	SOC004	NULL	NULL
S10010	SOC006	NULL	NULL
X10019	SOC004	NULL	NULL
Z10029	SOC003	2018-01-02	NULL

```
21 rows in set (0.007 sec)
```


Query 23: Create a view to keep track of society names with the total number of students enrolled in it.

```
mysql> CREATE VIEW SocietyEnrollment AS
-> SELECT s.SocName, COUNT(e.RollNo) AS TotalEnrolled
-> FROM SOCIETY s
-> LEFT JOIN ENROLLMENT e ON s.SocID = e.SID
-> GROUP BY s.SocID;
Query OK, 0 rows affected (0.161 sec)
```

Query 24: Find student names enrolled in all the societies.

```
mysql> SELECT s.StudentName
-> FROM STUDENT s
-> JOIN ENROLLMENT e ON s.RollNo = e.RollNo
-> GROUP BY s.RollNo
-> HAVING COUNT(DISTINCT e.SID) = (SELECT COUNT(*) FROM SOCIETY);
Empty set (0.243 sec)
```

Query 25: Count the number of societies with more than 5 students enrolled in it

```
mysql> SELECT COUNT(*) AS CountSocieties
-> FROM (
-> SELECT SID
-> FROM ENROLLMENT
-> GROUP BY SID
-> HAVING COUNT(*) > 5
-> ) AS temp;
+-----+
| CountSocieties |
+-----+
| 0 |
+-----+
1 row in set (0.029 sec)
```

Query 26: Add column Mobile number in student table with default value '9999999999'

```
mysql> ALTER TABLE STUDENT
-> ADD Mobile VARCHAR(10) DEFAULT '9999999999';
Query OK, 0 rows affected (0.323 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> select * from student;
+-----+-----+-----+-----+-----+
| RollNo | StudentName | Course | DOB | Mobile |
+-----+-----+-----+-----+-----+
| A10089 | AMIT DHOUDIYAL | CHEMISTRY | 2006-10-10 | 9999999999 |
| K10045 | RUPESH KUMAR | COMP SCI | 2006-03-11 | 9999999999 |
| L50007 | SOMU YADAV | PHYSICS | 2006-04-23 | 9999999999 |
| S10001 | KRISHNA GUPTA | COMP SCI | 2006-08-20 | 9999999999 |
| S10002 | PRINCE CHOUDHARY | CHEMISTRY | 2005-12-24 | 9999999999 |
| S10003 | KRISH KASHYAP | PHYSICS | 2006-11-09 | 9999999999 |
| S10004 | LOVELEEN KAPIL | COMP SCI | 2006-09-11 | 9999999999 |
| S10005 | MANDEEP CHOUDHARY | CHEMISTRY | 2006-02-14 | 9999999999 |
| S10006 | HARIT SHARMA | MATHS | 2006-03-16 | 9999999999 |
| S10007 | KRISH SALARIA | COMP SCI | 2006-08-22 | 9999999999 |
| S10008 | KRRISH RABUGOTRA | CHEMISTRY | 2006-02-17 | 9999999999 |
| S10009 | HARSHDEEP | PHYSICS | 2006-05-04 | 9999999999 |
| S10010 | YUVRAJ KAPIL | MATHS | 2006-12-24 | 9999999999 |
| X10019 | KAPIL BHAN | COMP SCI | 2005-12-17 | 9999999999 |
| Z10029 | HARSHVARDHAN MEENA | CHEMISTRY | 2004-10-11 | 9999999999 |
+-----+-----+-----+-----+-----+
15 rows in set (0.021 sec)
```

Query 27: Find the total number of students whose age is > 20 years.

```
mysql> SELECT COUNT(*) AS TotalStudents
-> FROM STUDENT
-> WHERE TIMESTAMPDIFF(YEAR, DOB, CURDATE()) > 20;
+-----+
| TotalStudents |
+-----+
|          1 |
+-----+
1 row in set (0.025 sec)
```

Query 28: Find names of students who were born in 2001 and are enrolled in at least one society.

```
mysql> SELECT DISTINCT s.StudentName
-> FROM STUDENT s
-> JOIN ENROLLMENT e ON s.RollNo = e.RollNo
-> WHERE YEAR(s.DOB) = 2001;
Empty set (0.048 sec)
```

Query 29: Count all societies whose name starts with 'S' and ends with 't' and at least 5 students are enrolled in the society.

```
mysql> SELECT s.SocName
-> FROM SOCIETY s
-> JOIN ENROLLMENT e ON s.SocID = e.SID
-> WHERE s.SocName LIKE 'S%t'
-> GROUP BY s.SocID
-> HAVING COUNT(*) >= 5;
Empty set (0.010 sec)
```

Query 30: Display the following information:

Society name	Mentor name	Total Capacity	Total Enrolled	Unfilled Seats
--------------	-------------	----------------	----------------	----------------

```
mysql> SELECT s.SocName,
-> s.MentorName,
-> s.TotalSeats AS TotalCapacity,
-> COUNT(e.RollNo) AS TotalEnrolled,
-> s.TotalSeats - COUNT(e.RollNo) AS UnfilledSeats
-> FROM SOCIETY s
-> LEFT JOIN ENROLLMENT e ON s.SocID = e.SID
-> GROUP BY s.SocID;
+-----+-----+-----+-----+-----+
| SocName | MentorName | TotalCapacity | TotalEnrolled | UnfilledSeats |
+-----+-----+-----+-----+-----+
| Debating | Pardeep Sharma | 11 | 3 | 8 |
| Dancing | Ajay Verma | 9 | 3 | 6 |
| Sashakt | Asha Baloria | 13 | 3 | 10 |
| NSS | Sudhir Kataria | 17 | 3 | 14 |
| Coding | Bharat Bhushan | 22 | 4 | 18 |
| Drama | Aman Gupta | 6 | 1 | 5 |
| Sports | Gautam Gambhir | 28 | 4 | 24 |
| Music | Sanjay Gupta | 11 | 0 | 11 |
+-----+-----+-----+-----+-----+
8 rows in set (0.012 sec)
```

II. Do the following database administration commands: Create user, create role, grant privileges to a role, revoke privileges from a role, create index

```
mysql> CREATE USER 'student1'@'localhost' IDENTIFIED BY 'password123';
Query OK, 0 rows affected (1.037 sec)

mysql> CREATE ROLE 'society_manager';
Query OK, 0 rows affected (0.069 sec)

mysql> GRANT SELECT, INSERT, UPDATE
      -> ON college_db.*
      -> TO 'society_manager';
Query OK, 0 rows affected (0.144 sec)

mysql> GRANT 'society_manager' TO 'student1'@'localhost';
Query OK, 0 rows affected (0.080 sec)

mysql> SET DEFAULT ROLE 'society_manager' TO 'student1'@'localhost';
Query OK, 0 rows affected (0.064 sec)

mysql> REVOKE INSERT
      -> ON college_db.*
      -> FROM 'society_manager';
Query OK, 0 rows affected (0.071 sec)

mysql> CREATE INDEX idx_student_name
      -> ON STUDENT(StudentName);
Query OK, 0 rows affected (0.971 sec)
Records: 0 Duplicates: 0 Warnings: 0
```